Vacuum 236 (2025) 114161

Contents lists available at ScienceDirect

Vacuum

journal homepage: www.elsevier.com/locate/vacuum

Preparation, microstructure and properties of FeCrAlTiSi high entropy alloy coatings using the combined technique of magnetron sputtering with multiarc ion plating

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ARTICLE INFO

Handling Editor: Prof. L.G. Hultman

Keywords:
High-entropy alloy coatings
FeCrAlTiSi
Arc ion plating
Magnetron sputtering
Mechanical property
Corrosion resistance

ABSTRACT

In this work, FeCrAlTiSi high-entropy alloy coatings (HEACs) were prepared using magnetron sputtering (MS) under various substrate biases with a Ti interlayer pre-deposited through multiarc ion plating (M-AIP) to enhance the coating/substrate adhesion strength. The morphology, microstructure, elemental and phase composition, nanohardness, Young's modulus, adhesion strength and corrosion resistance were characterized using SEM, XRD, TEM, XPS, nanoindenter, scratch tester and electrochemical workstation. Results indicated that all FeCrAlTiSi coatings were in an amorphous state. Nanohardness and Young's modulus had a strong substrate bias dependence, reaching the maximum of 16.04 GPa and 241.3 GPa, respectively, at −100 V bias. Meanwhile, FeCrAlSiTi coatings at −100 V exhibited superior corrosion resistance to that of 316L in 3.5 wt% NaCl aqueous solution, about 60 times improvement in terms of corrosion current density owing to the combined effect of homogeneous composition distribution, amorphous structure, and chemically stable oxide film predominantly composed of Al_2O_3, TiO_2 and SiO_2.

1. Introduction

It is still challenging to develop surface coating with superior comprehensive performance such as high hardness, better corrosion resistance, high antioxidation temperature, etc., to meet the ever-increasing demands driven by the continuous advancement of the automotive, marine, aviation, and other machinery industries [1–7]. Compared with the traditional alloy coatings with single-principal-component metal, high entropy alloy coatings (HEACs) with multiple principal components of equal or nearly equal molar ratios have excellent comprehensive properties, owing to high entropy, slow diffusion, cocktail effects, and lattice distortion effects [8–10]. As a result, HEACs have attracted increasing attention from researchers, with an emphasis focused on fabrication, structure, and properties.

Currently, various methods have been attempted to fabricate high-performance HEACs, including electro-spark deposition [11], laser cladding [10,12], plasma spraying [13,14], magnetron sputtering [15–17], multi-arc ion plating [18,19], and so on. These researches demonstrated that crystallite HEACs with fcc and/or bcc crystal structure, such as fcc $Fe_{3.5}Ni_{3.5}Cr_2MnAl_{0.7}$ coating [20], fcc $FeCoCrNiAl_{0.1}N$ coating [18], bcc TiTaNbZrMo coating [16], have remarkable nanohardness but with relatively limited corrosion resistance. In contrast, amorphous coatings have excellent corrosion resistance owing to lacking grain boundaries and homogeneous structure [12,21,22]. Unfortunately, to the best of our knowledge, amorphous structured HEACs have been barely researched to date.

Nowadays, multi-arc ion plating and magnetron sputtering are the two most widely used physical vapor deposition techniques in the industry to prepare the coatings. The magnetron sputtering technique has distinct advantages in preparing coating, such as low substrate temperature, fine grain coating, and precise control over coating thickness, but with weak adhesion strength to substrate and low deposition rate

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https://doi.org/10.1016/j.vacuum.2025.114161
Received 12 December 2024; Received in revised form 18 February 2025; Accepted 19 February 2025
Available online 20 February 2025
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[23–25]. In contrast, multi-arc ion plating is well known for its high ionization and deposition rate, but with a few sparse micron-sized particles in the coating as a result of congelation of the injected droplet during arc-discharging [26,27].

Herein, to take full advantage of the merits of M-AIP, MS, amorphous coating, and high-entropy alloy, amorphous FeCrAlTiSi high-entropy alloy coatings were designed and fabricated through an integrated M-AIP and MS technology. The effect of substrate bias on the microstructure, mechanical properties, and electrochemical corrosion behavior was investigated in detail.

2. Experimental details

The equiatomic FeCrAlTiSi HEA target (400 mm × 80 mm × 6 mm) was prepared using the vacuum hot-press sintering method, following the procedures reported in Ref. [28]. The Ti target (ϕ100mm × 40 mm) was prepared using vacuum arc melting. FeCrAlTiSi coatings were fabricated through our self-designed and manufactured vacuum ion plating equipment which integrated magnetron sputtering (MS) and multiarc ion plating (M-AIP), as schematized in Fig. 1.

Silicon (100) wafers and 316L steels were employed as substrates for structural characterization and properties measurement, respectively. Before coating deposition, the 316L substrate surface was sequentially ground with silicon carbide papers of grits 400, 800, 1500, and 2000, followed by mechanical polishing with diamond lapping pastes down to 1.5 μm. Thereafter, all substrates were ultrasonically cleaned and rinsed with acetone, ethanol, and deionized water for 15 min.

The substrates were placed on the substrate holders parallel to the targets, the target-to-substrate distance was maintained at 10 cm. To avoid possible residual gas contamination, the sputter chamber was pumped to 5×10^{-3} Pa at first, followed by baking at 200 °C for about 1 h and cooling naturally to about room temperature subsequently while keeping pumping. Prior to the experiment, the base pressure of the sputter chamber was 2×10^{-3} Pa. The argon flow rate was fixed at 50 sccm, and the working pressure was held at 0.5 Pa. The Ti target was pre-sputtered for 15 min at an arc current of 70 A and a bias voltage of −400 V. In addition, the FeCrAlTiSi target was pre-sputtered for 15 min at the power of 600 W and a bias of −400 V. To enhance the adhesion of the coating to the substrate, a Ti interlayer was first deposited on the substrates using M-AIP, with a substrate bias of −100 V and arc current of 70 A for 3 min, and then FeCrAlTiSi coatings were deposited subsequently using magnetron sputtering technique in the power of 400 W under different biases from 0 to −200 V for 3 h.

The surface and cross-sectional morphology of the coatings were characterized by field emission scanning electron microscopy (FESEM, Sirion 200, FEI). Elemental distribution and content were determined by energy dispersive spectroscopy (EDS) in conjunction with FESEM. Microstructural information was obtained through field emission transmission electron microscopy (TEM, JEOL, JEM-2010) operated at 200 keV. The crystal structure was characterized by X-ray diffraction (XRD, Philips X'pert PROMPD), using Cu Kα radiation (α = 0.154 nm) with a scan range of 20°–80° in 0.06° steps with 0.88 s exposure. In order to exclude the influence of the possible residual water vapor and/or hydrogen, our coatings prepared at −100V bias were analyzed using Raman spectroscopy (Raman, Renishaw, inVa Reflex) with a green laser equipped (wavelength 532 nm) and Fourier Transform Infrared Spectrometer (FTIR, Thermo Scientific, iS50R FT-IR). The chemical bonding state and elemental composition were analyzed using X-ray photoelectron spectroscopy (XPS, Kratos Axis supra$^+$) with Al Kα (characteristic energy: 1.4866 KeV) radiation at a pass energy of 40 eV. The instrument used a power of 150 W and an analytical area of 300 μm × 700 μm. Before XPS measurements, the sample surface was etched using an Ar^+ ion beam (1 keV) with an etching duration of 3 min until the elemental composition reached a steady state to remove contaminants due to air exposure, meanwhile, without severe surface damage. The work functions of the coatings were measured by ultraviolet photoelectron spectroscopy (UPS, Kratos Axis supra$^+$). All XPS spectra were fitted using the Avantage software.

Nanohardness and Young's modulus of the coatings were measured with a nano-indenter (MTS NANO Indenter® G200) in continuous stiffness measurement (CSM) mode at room temperature. In measurements, the penetration depth was kept at about 1/7–1/10 of the coating thickness to avoid the effects of substrate and the surface roughness of coatings. The data were averaged from 10 tests per sample to improve measurement accuracy.

The residual stress of the coatings was measured using a film stress tester (FST 1000, SuPro) based on the Stoney equation:

$$\sigma = \frac{E_s \cdot t_s^2}{6(1-\nu_s)t_f}\left(\frac{1}{R_1}-\frac{1}{R_0}\right) \quad (1)$$

where E_s, t_s and ν_s are the Young's modulus, thickness and Poisson's ratio of substrate, and t_f represents the thickness of coating. R_1 and R_0 are the curvature radius of substrate after and before coating preparation.

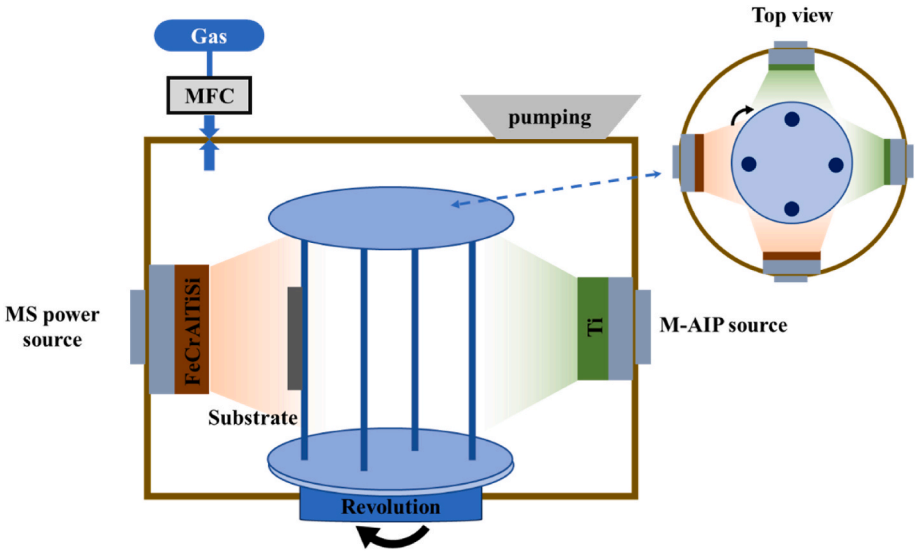


Fig. 1. Schematic diagram of self-designed and manufactured vacuum ion plating equipment with two magnetron sputtering targets and two multiarc ion plating sources aligned alternatively. The substrate holder can rotate at a rate of 0–15 rpm.

The adhesion strength was assessed using a scratch tester (Anton Paar) with a 0.2 mm diameter, 120° taper diamond indenter. A diamond indenter was dragged across the coating surface through a distance of 3 mm at 0.1 mm/s, with a gradually increasing load from 0 to 30 N.

Electrochemical corrosion behavior was performed using an electrochemical workstation (CHI660E) in 3.5 wt% NaCl aqueous solution at room temperature. The tests were carried out using a three-electrode electrochemical cell with a saturated calomel electrode (SCE) as the reference electrode, a platinum sheet as the auxiliary electrode, and the test samples as the working electrode. The potentiodynamic polarization experiments were scanned at 1 mV/s with electrode potentials scanning from −0.8 to 1 mV (vs SCE). The EIS experiments were performed at the open circuit potential (OCP) with a sinusoidal potential amplitude of 10 mV over frequencies ranging from 10^5 Hz to 0.01 Hz. To confirm the reproducibility of the data, the electrochemical tests were repeated at least three times.

3. Results and discussion

3.1. Microstructure

Fig. 2 illustrates the surface and cross-sectional SEM images of FeCrAlTiSi coatings grown at different biases, along with the statistical distribution of column width and particle size. The thickness of the coatings deposited at the bias of 0 V, −50 V, −100 V, −150 V, and −200 V is 2.95 μm, 3.98 μm, 4.92 μm, 4.36 μm, and 3.45 μm, respectively, showing an initial increase and then decrease trend with the maximum value of 4.92 μm at −100 V, which could be understood in terms of the target atom/ion transport from target to substrate during sputtering process. As the bias voltage is increased from 0 to −100 V, the higher the bias is the larger the amount of energetic target ions deposited on the substrate to form coating before their kinetic energy is reduced to zero due to cascading collision with other species (atom, ions in the plasma). Thereafter, the decrease of coating thickness with bias increasing is due to the predominant re-sputtering effect of substrate bias on coating [29–31]. It also can be observed that all coatings exhibited a typical columnar structure with column width of 174.8–260.8 nm and granular surface morphology consisted of uniformly sized nanoparticles of

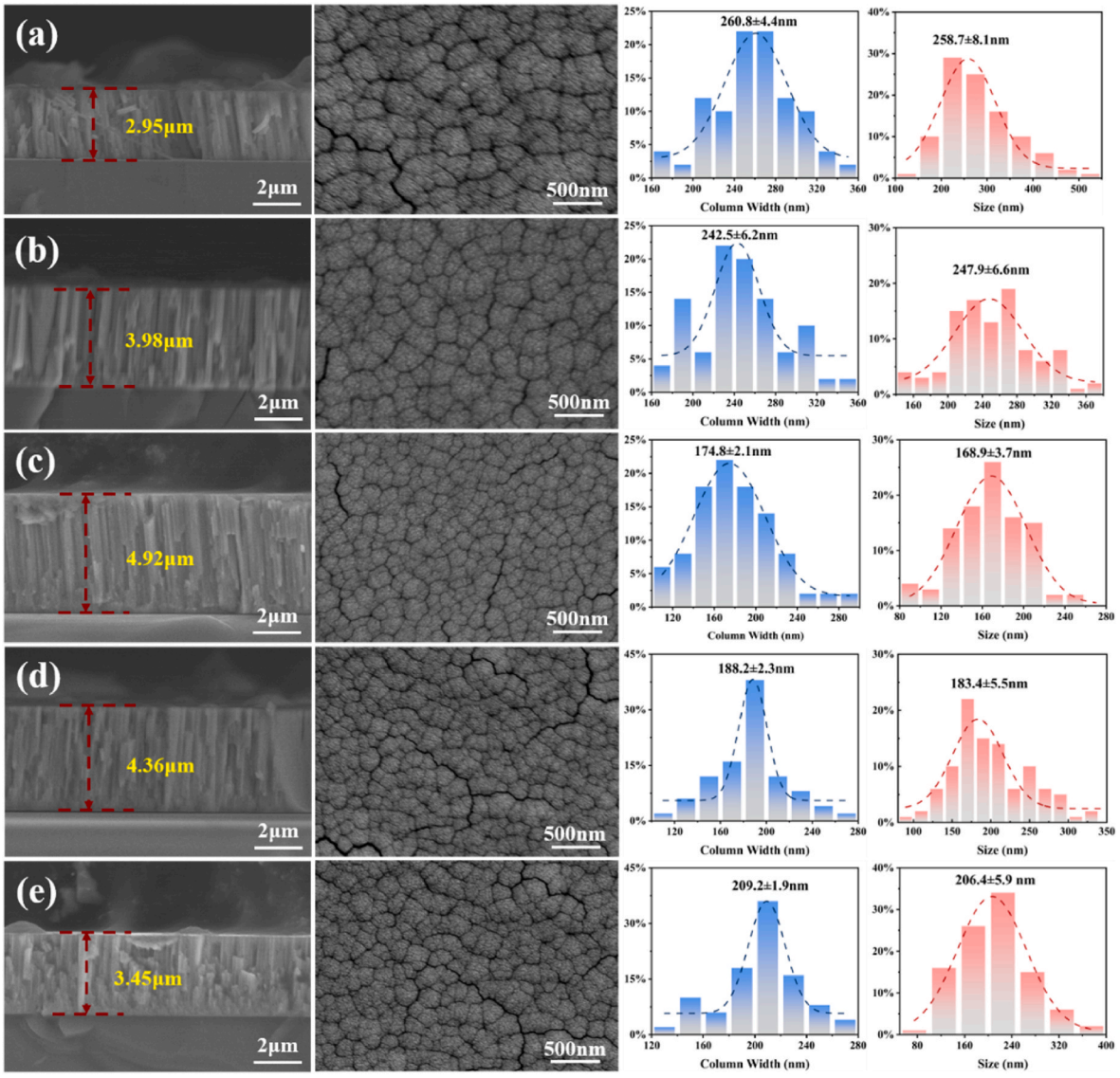


Fig. 2. The cross-sectional, surface, column width, and particle size distribution map of the FeCrAlTiSi coatings deposited at the bias of (a) 0 V, (b) − 50 V, (c) − 100 V, (d) − 150 V and (e) − 200 V.

168.9–258.7 nm. Both the column width and the particle size reach the minimum of 174.8 nm and 168.9 nm, respectively, at the substrate bias of −100 V, demonstrating the formation of the densest and finest coating.

Fig. 3 presents the elemental composition of FeCrAlTiSi coatings deposited at different biases and the element mapping at −100 V. It can be seen that all coatings consist of Fe, Cr, Al, Ti, and Si, with an atomic ratio close to the customized target. The substrate bias has no obvious influence on the elemental composition. All these elements are uniformly distributed into the coating, suggesting that the coatings are homogeneously deposited. Fig. 4 presents the FTIR and Raman spectra of FeCrAlTiSi coating at −100 V bias. No discernible peaks of C-H, Si-H, O-H or Si-O-H bonds appear in both FTIR and Raman spectra, which implies that the influence of the residual water vapor and hydrogen could be neglected. The FTIR absorption peaks of Ti-O, Si-O and Al-O bonds are observed clearly, indicating the existence of Ti, Si and Al oxides, which might be due to the impurities in the working gas, as well as the presence of oxide phases in the target composition. The oxygen content of the coatings prepared at different biases remained essentially constant, with a slight decrease with bias increasing, which could be explained in terms of the resputtering of the coating. As the substrate bias is increased, Ar^+ ions bombard the coating surface, oxygen atoms with relatively lighter atom weight are more likely to be resputtered from the coating surface than that of metal atoms, and therefore the oxygen content is decreased slightly [32]. It should be pointed out that in all coatings, oxygen content was well controlled below 5 at.%, and therefore was excluded in the element composition of the coatings. Fig. 5 shows the cross-sectional TEM images and the corresponding EDS mapping of the FeCrAlTiSi coating prepared under −100 V. The obvious columnar structure can be observed with all elements uniformly distributed in the coating, without any oxygen segregation in layer interfaces and columnar boundaries within the layer. Fig. 6 (a) is the XRD pattern of the FeCrAlTiSi coatings prepared under different biases. All coatings exhibit a single broad diffraction peak at about 45° and noticeable background noise, without distinct crystalline peaks. It indicates that the FeCrAlTiSi coatings have an amorphous structure [22], which could be further confirmed by the HRTEM and SAED pattern in Fig. 6(b) and (c). No crystalline phases or nanograins can be observed in the HRTEM images. The fast Fourier transform (FFT) images show the halo ring patterns, with a maze and random structure that can be seen in the inverse fast Fourier transform (IFFT) images. The SAED pattern exhibits two dispersive electron diffraction rings. These characteristics indicate its amorphous nature, which is consistent with the XRD result.

Theoretically, the phase formation of the high entropy alloys can be predicted using the classical Hume-Rothery rule regarding alloy formation and the entropy-enthalpy balance in the system [31,33–35], which can be expressed by the following equations (Eqs. (2)-(6)):

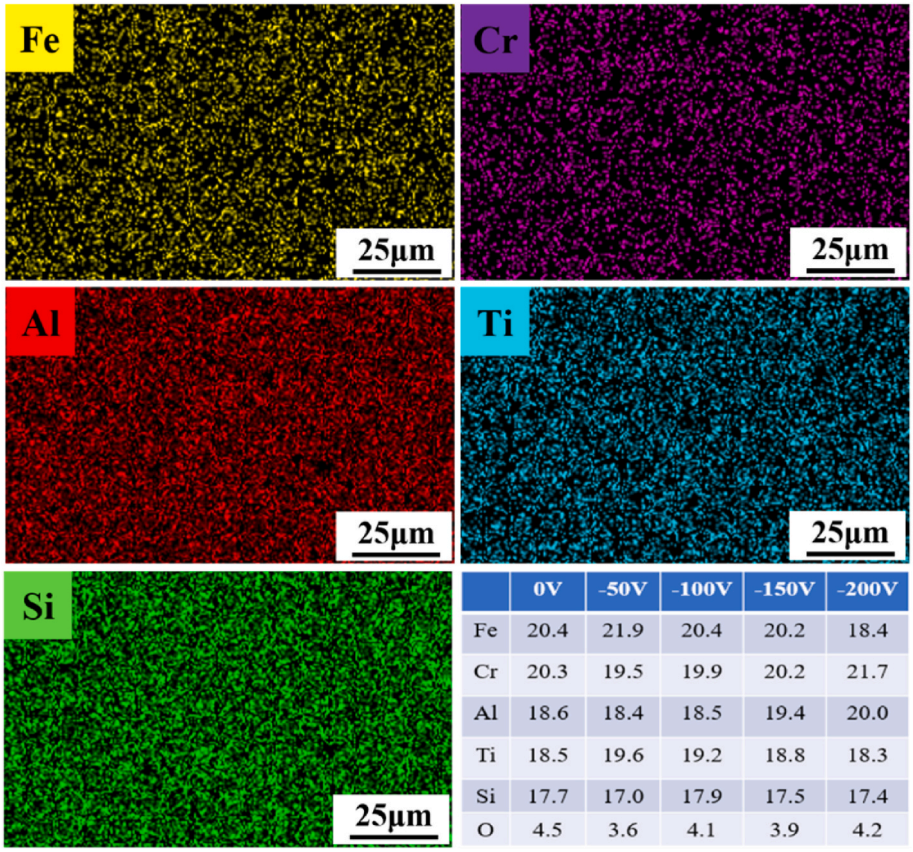
$$\delta = \sqrt{\sum_{i=1}^{n} c_i (1 - r_i/\bar{r})^2} \tag{2}$$

$$\Delta H_{mix} = \sum_{i=1 j\neq i}^{N} 4\Delta H_{AB}^{mix} c_i c_j \tag{3}$$

$$\Delta S_{mix} = -R \sum_{i=1 j\neq i}^{N} c_i \ln c_i \tag{4}$$

$$\Omega = \frac{\Delta S_{mix} \cdot T_m}{|\Delta H_{mix}|} \tag{5}$$

$$T_m = \sum_{i=1}^{n} c_i (T_m)_i \tag{6}$$



	0V	-50V	-100V	-150V	-200V
Fe	20.4	21.9	20.4	20.2	18.4
Cr	20.3	19.5	19.9	20.2	21.7
Al	18.6	18.4	18.5	19.4	20.0
Ti	18.5	19.6	19.2	18.8	18.3
Si	17.7	17.0	17.9	17.5	17.4
O	4.5	3.6	4.1	3.9	4.2

Fig. 3. EDS element distribution of the FeCrAlTiSi coating deposited at −100 V and elemental composition of coatings deposited at different biases.

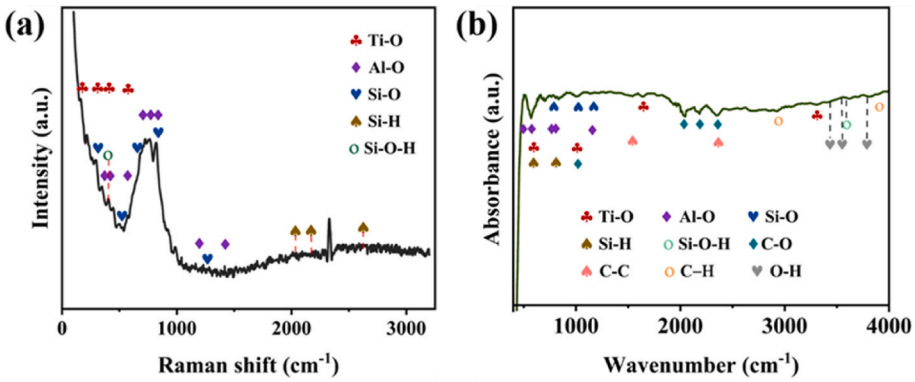


Fig. 4. Raman (a) and FTIR (b) spectra of the FeCrAlTiSi coating prepared under −100 V.

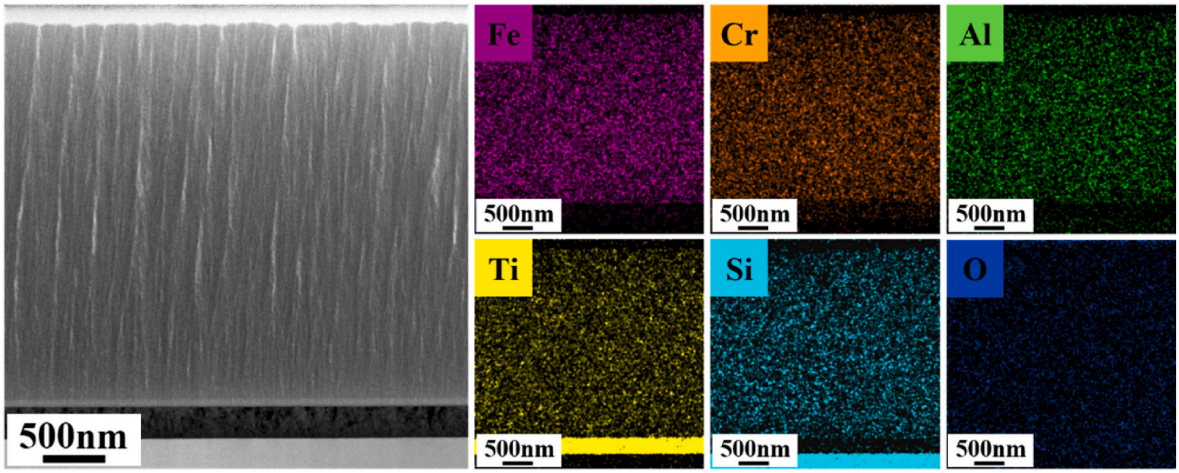


Fig. 5. Cross-sectional TEM images of the FeCrAlTiSi coating prepared under −100 V and the corresponding EDS mapping.

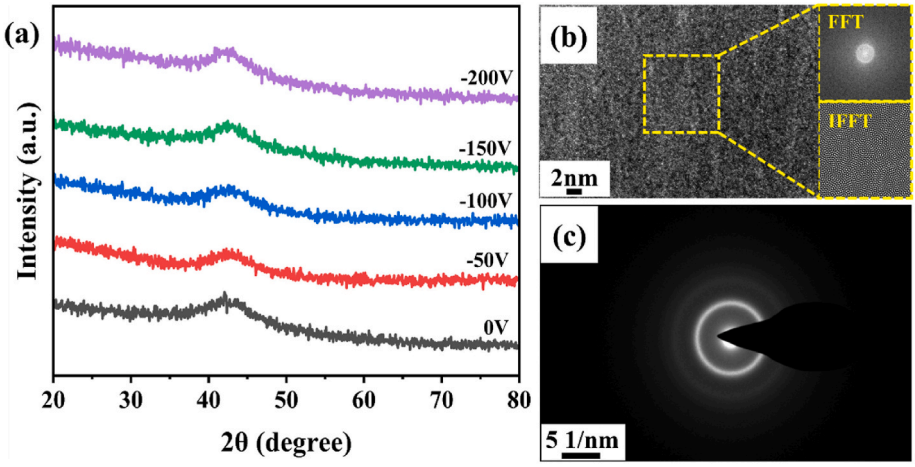


Fig. 6. (a) The XRD patterns of the FeCrAlTiSi coatings prepared under different biases, (b) cross-section HETEM morphology with corresponding fast Fourier transform (FFT) and inverse fast Fourier transform (IFFT) patterns and (c) SAED pattern of FeCrAlTiSi coating deposited at −100 V.

where δ, ΔH_{mix}, ΔS_{mix} and Ω represent atomic radius mismatch, enthalpy of mixing, entropy of mixing, and Yang-Zhang parameter, respectively. c_i, r_i and $(T_m)_i$ are the atomic fraction, radius, and melting temperature of the ith component. $\bar{r}$, ΔH_{AB}^{mix}, T_m and R denote the average atomic radius, the mixing enthalpy of binary atomic pairs, the melting temperature, and the universal gas constant (R = 8.314 J/(mol·K)), respectively. It has been proved that amorphous alloys tend to form in the case of $\delta \geq 9$, $-49 \leq \Delta H_{mix} \leq -5.5$ kJ/mol and $7 \leq \Delta S_{mix} \leq 16$ J/(K·mol) [33] or $\Omega \leq 1.1$ and $\delta \geq 6.6$ [35]. In our case, the δ, Ω, ΔS_{mix}, ΔH_{mix} of FeCrAlTiSi coatings prepared under different biases were calculated and listed in Tables 1 and 2. As seen that ΔH_{mix} of −36.51 ~ −35.96 is between −49 and −5.5, δ of 8.95–9.01 ≥ 6.6, ΔS_{mix} of 13.35–13.37 is between 7 and 16, Ω of 0.53–0.54 ≤ 1.1, which demonstrates that our coatings tend to form amorphous coatings, in

Table 1
The values of three-phase prediction parameters $\bar{r}$, δ, Ω, ΔH_{mix} and ΔS_{mix} of the FeCrAlTiSi coatings.

Bias(V)	$\bar{r}$ (Å)	δ(%)	Ω	ΔH_{mix}(kJ/mol)	ΔS_{mix} (J/K·mol)
0	1.306	8.97	0.54	−35.96	13.37
−50	1.307	8.95	0.53	−36.28	13.35
−100	1.306	9.01	0.53	−36.51	13.37
−150	1.307	8.98	0.53	−36.00	13.36
−200	1.307	8.97	0.54	−35.60	13.35
Target	**1.307**	**9.12**	**0.51**	**−37.28**	**13.38**

Table 2
The values of ΔH^{AB}_{mix} (kJ/mol) for atomic pairs in the FeCrAlTiSi coatings.

	Fe	Cr	Al	Si	Ti
Fe	/	−1	−11	−35	−17
Cr	−1	/	−10	−37	−7
Al	−11	−10	/	−19	−30
Si	−35	−37	−19	/	−66
Ti	−17	−7	−30	−66	/

agreement with our XRD and HRTEM results.

In order to clarify the bonding state of FeCrAlSiTi coatings, XPS analysis was conducted, with the detailed spectra shown in Fig. 7. Regarding the charge displacement calibration, G. Greczynski et al. [36–41] demonstrated that aligning the spectra to the C 1s peak of adventitious carbon was not dependable as it might lead to unphysical results and that the binding energy of the C 1s peak of the indeterminate carbon was strongly associated with changes in the sample work function (φ_{SA}), with the sum of the C1s peak binding energy and φ_{SA} being constantly 289.58 ± 0.12 eV. The work function deposited at the bias of −100 V was determined using UPS as 4.71 eV to calibrate the XPS spectra. Consequently, the C1s peak binding energy of the coating was measured as 284.87 eV. The XPS spectra of other elements were calibrated using the C1s binding energy as a reference. The XPS survey spectra shown in Fig. 7 (a) implied the presence of seven elements: C, O, Ar, Fe, Cr, Al, Ti and Si. The presence of C might be attributed to contamination of the sample before or during the testing process, as well as background contamination in the equipment. The O identified in the spectra may be due to small amounts of residual O_2 in the chamber. Besides, Ar may originate from the Ar species during growth and/or etching. Due to the charge difference caused by the altered electrical properties in the argon-influenced layer, the pristine peaks may be shifted to higher binding energies after argon ion etching, along with the

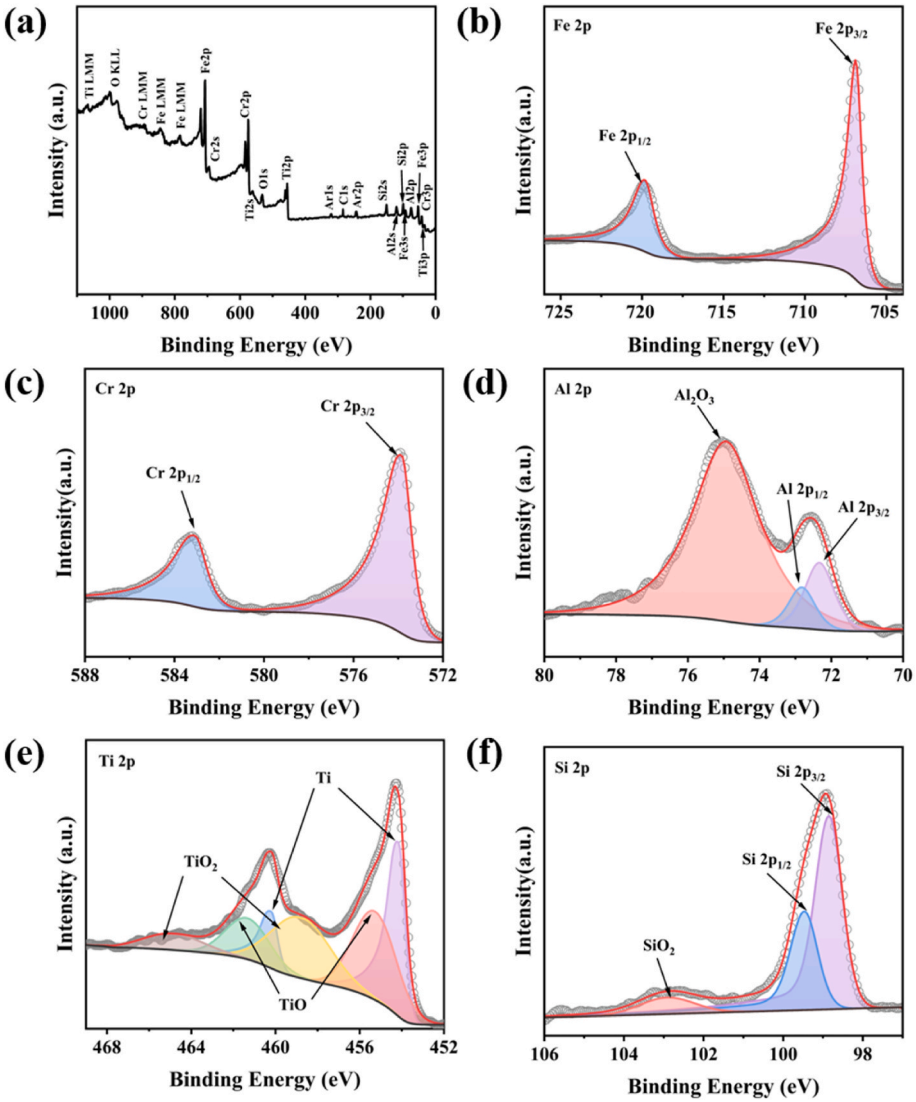


Fig. 7. XPS metal peaks of FeCrAlTiSi coating deposited at −100 V: (a) survey spectra, (b) Fe 2p, (c) Cr 2p, (d) Al 2p, (e) Ti 2p, and (f) Si 2p.

broadening of the metal line generation [42]. The Fe 2p spectrum presents two peaks centered at the binding energy of 706.87 eV and 719.83 eV, with a peak separation of 13.14 eV, corresponding to the Fe2p$^{3/2}$ and Fe2p$^{1/2}$ of metallic Fe [43]. In the case of the Cr 2p spectrum, two peaks center at 573.84 eV and 583.09 eV, corresponding to 2p$^{3/2}$ and 2p$^{1/2}$ of metallic Cr [43,44]. The Al 2p spectrum could be decomposed into three peaks at 72.33 eV, 72.82 eV, and 74.93 eV, in agreement with 2p3/2 and 2p1/2 of Al and Al 2p of Al_2O_3 [45]. The Ti 2p spectrum could be decomposed into four peaks centered at 454.16 eV, 460.21 eV, 455.29 eV, 461.34 eV, 458.67 eV, and 464.80 eV, which could be assigned to 2p$^{3/2}$ and 2p$^{1/2}$ of metallic Ti, TiO and TiO_2 [46]. The Si 2p spectrum reveals three peaks centered at 98.85 eV, 99.47 eV and 102.91 eV, corresponding to 2p$^{3/2}$ and 2p$^{1/2}$ of Si and Si 2p of SiO_2 [47]. These results suggest the preferential surface oxidation of Ti, Si, and Al owing to their higher negative heat of formation of oxides, which agrees with the theoretical simulation results in the literature [48].

3.2. Mechanical properties

The variation of hardness (H), Young's modulus (E), residual stress, H/E and H^3/E^2, and load-displacement curves of FeCrAlTiSi coating with substrate bias is shown in Fig. 8. As the bias is increased from 0 V to −200 V, both the hardness and Young's modulus initially increase and then decrease, reaching the maximum value of 16.04 GPa and 241.3 GPa, respectively, at −100 V. They are significantly higher than the hardness (~3.68 GPa) and Young's modulus (~133.3 GPa) of 316L substrate, mainly owing to the solid solution strengthening effect arising from the large atomic size mismatch in FeCrAlSiTi HEA coatings. After applying bias, the residual stress of coatings increases first and then decreases, with the highest value of ~0.36 GPa at − 100 V, which is in consistence with the change of hardness and Young's modulus. The observed variation in coating hardness with substrate bias could be attributed to microstructure densification and the residual internal stresses [18,31,49,50]. During deposition, applying a negative substrate bias induces positive ions to bombard coatings, leading to residual stress in the coatings. However, enhanced ion bombardment also improves the mobility and diffusivity of the adsorbed atoms, promoting the fusion of the internal defects and balancing some residual stresses. The competing relationship between the formation and fusion of the internal defects produces a difference in microstructure and affects the properties of the coatings.

The toughness of coatings can be evaluated using the ratio of H/E and H^3/E^2. Generally, a higher H/E ratio correlates with improved toughness, while a higher H^3/E^2 ratio implies greater load-bearing capacity without yielding. Fig. 8 (b) depicts the variation of H/E ratio, H^3/E^2 with substrate bias. It can be seen that H/E and H^3/E^2 almost follow the same variation trend as that of hardness and Young's modulus in Fig. 8 (a), reaching the maximum at −100 V. These results demonstrated that our FeCrAlSiTi coatings had the optimum comprehensive mechanical properties at −100 V, which were much better than those of bcc-FeCrAl, bcc-NbTaTiVZr,bcc-VNbMoTaWCr, fcc-FeCoNiCu, fcc-CrNbTiMoZr, fcc-AlCoCrFeNiTi, fcc/bcc-AlCoCrCuFeNi, nc/a-AlCrNbTiSi high entropy alloy coatings [31,51–57], as shown in Fig. 9, which could be attributed to the amorphous nature and the densely compact microstructure of our coatings. In contrast to crystallized high entropy alloy coatings, amorphous coatings lack grain boundaries, dislocations, and laminar faults. Therefore, when subjected to stress, the atoms in the amorphous coating move collectively, exhibiting shear deformation, which makes it necessary for the amorphous coating to have higher stress to induce deformation and result in increased hardness. Additionally, the diffusion coefficient of atoms in high-entropy alloys is lower than in reference metals, leading to slower atomic diffusion, which hinders shear band formation and further enhances the coating's hardness [58].

Fig. 10 shows the scratch test results of FeCrAlTiSi coatings prepared under different biases. It can be seen that the coatings without bias or

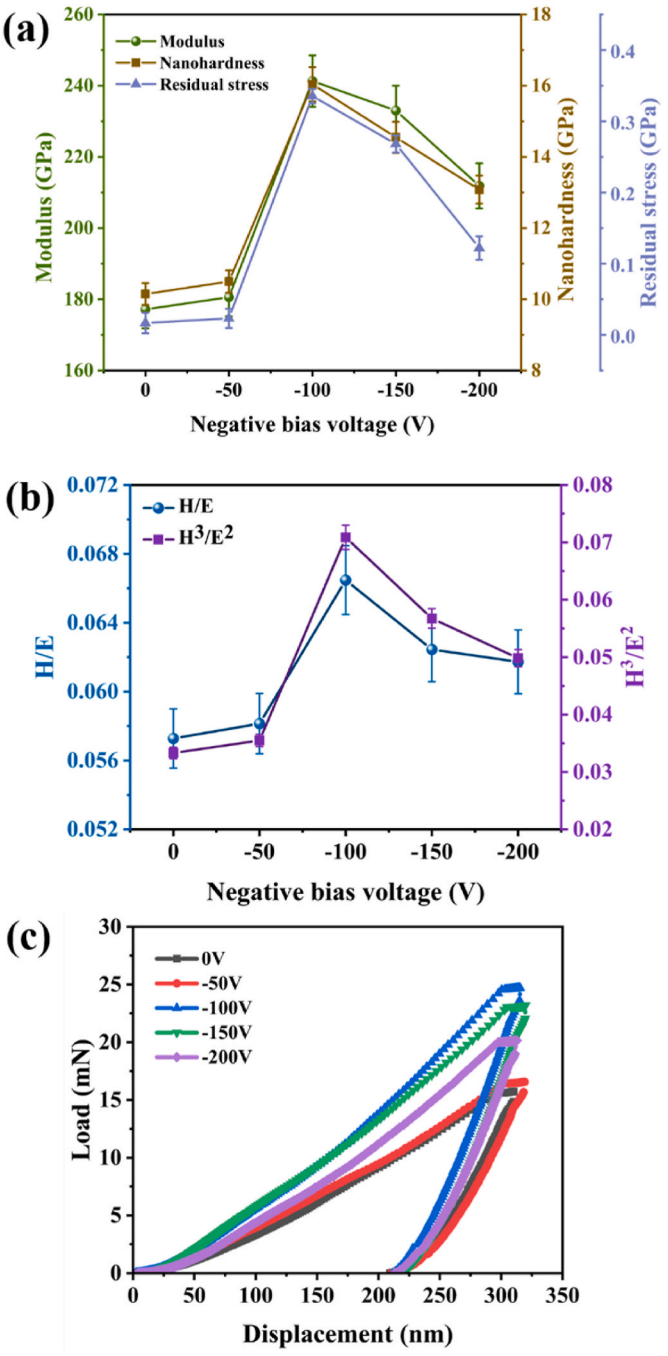


Fig. 8. Variation of (a) hardness, elastic modulus and residual stress, (b) H/E and H^3/E^2 with the bias of the FeCrAlTiSi coatings, and (c) the load-displacement curves.

with insufficient bias have poor scratch toughness and spalling occurs during the initial testing phase. The coating under −100 V bias exhibits the highest critical load of 14.54 N, with only a few flaking and cracks observed at the edge of sliding tracks. As the bias increases, the critical load decreases along with more severe flaking. The maximum adhesion strength of the coating to the substrate at −100 V can be attributed to the optimum hardness, H/E, and H^3/E^2 [21,31].

3.3. Corrosion behavior

Fig.11 (a) displays the potentiodynamic polarization plots of 316 L substrate and FeCrAlTiSi coatings, which were prepared under various

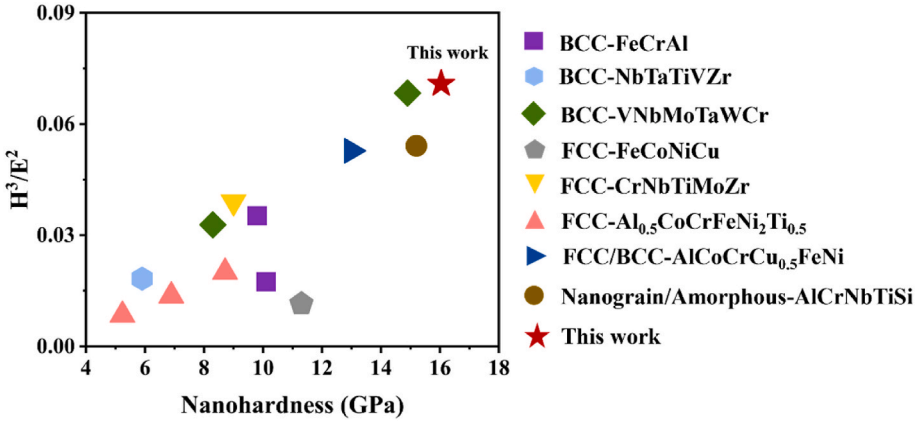


Fig. 9. Comparison of mechanical properties of FeCrAlTiSi coatings and other high-entropy alloy coatings [31,51–57].

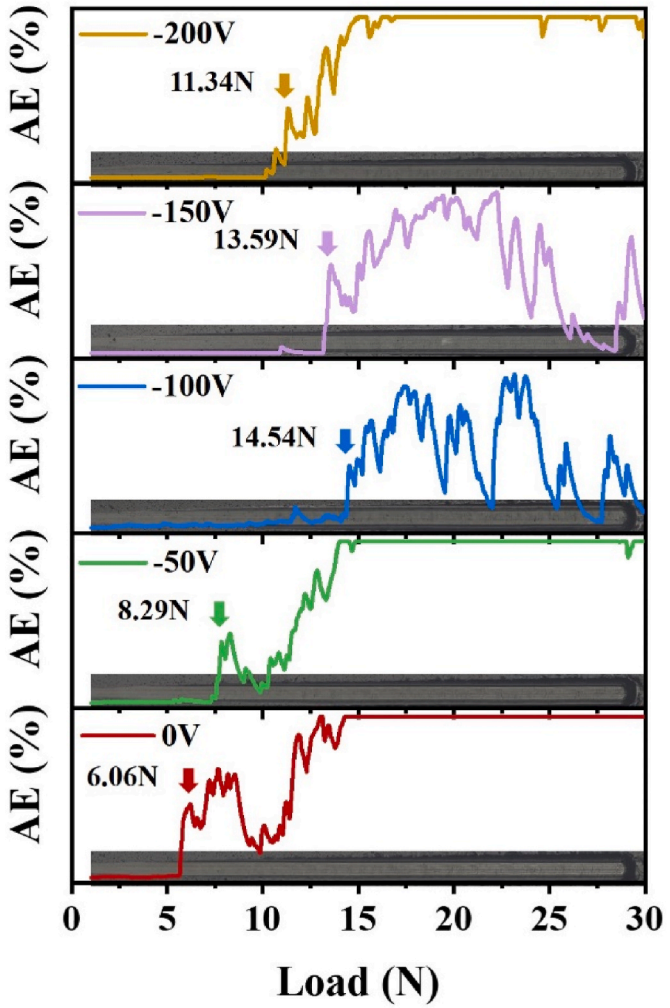


Fig. 10. Acoustic emission signals and scratch tracks of FeCrAlTiSi coatings prepared under different biases, and their critical loads.

substrate biases and immersed in a 3.5 wt% NaCl aqueous solution. It can be observed that both 316L and FeCrAlTiSi coatings are spontaneously passive with regions of primary and secondary passivity perhaps associated with great or lesser oxidation of specific elements at certain potentials, changes in oxidation state, or both. The passive current density (I_p) of HEA coatings was much smaller than that of 316L, which demonstrated that HEA coatings had better corrosion resistance than 316L substrate. To quantitatively evaluate the corrosion resistance, the corrosion potential (E_{corr}), corrosion current density (I_{corr}), and polarization resistance (R_p) were calculated and listed in Table 3 through Tafel extrapolation and Stern-Geary Equation [55], expressed as follows:

$$I_{corr} = \frac{\beta_a \cdot \beta_c}{2.3R_p \cdot (\beta_a + \beta_c)} \quad (7)$$

where β_a and β_c are the slopes of the anodic and cathodic curves, respectively. It can be seen that the I_{corr} of FeCrAlTiSi coatings is 0.122, 0.043, 0.025, 0.037, 0.225 μA·cm^{-2} at the substrate bias of 0, -50, −100, −150 and −200V, about one or two orders of magnitude lower than that of bare 316L, respectively. The E_{corr} of FeCrAlTiSi coatings is much higher than bare 316L. Correspondingly, the Rp is 289.65, 661.06, 909.29, 701.46, 150.01 kΩ, about several tens of times that of 316L. The corrosion current density of FeCrAlTiSi coatings increases first and then decreases, with the minimum value of ~0.025 μA·cm^{-2} at −100 V. The variation of corrosion current density with bias is mainly related to the compactness and thicknesses of the coatings. The minimum corrosion current density of ~0.025 μA·cm^{-2} at −100 V was owing to the narrowest columnar width, the smallest surface particles and the largest film thickness.

Electrochemical impedance spectroscopy of FeCrAlTiSi coatings and the bare 316L were measured after immersing in 3.5 wt% NaCl aqueous solution for 2 h. The Nyquist plots of both 316L and FeCrAlTiSi coatings exhibit a a depressed semicircle and a one-quarter circle, respectively, as shown in Fig. 11 (b). It is known that the diameter of the semicircle in the Nyquist plot represents charge transfer resistance and therefore is proportional to the corrosion resistance [12,59]. The larger diameter for FeCrAlTiSi HEA coatings indicates its corrosion resistance is superior to that of 316 L, notably in the case of −100V. To address capacitive dispersion of the double layer due to the non-uniform distribution of currents and/or oxide defects, a constant phase element (CPE) is used to model the non-ideal capacitance and its impedance is given by equation (8) [60]:

$$Z_{CPE} = \frac{1}{Y_0(jw)^n} \quad (8)$$

where Y_o is the capacitance, ω is the angular frequency, and n is the dispersion coefficient, usually describing the system's inhomogeneity in the range of 0–1. The corresponding parameters derived from the fitting process are summarized in Table 4. Specifically, R_s is the solution resistance; CPE_{op} denotes the capacitance associated with both the corrosion product and the corrosion solution; R_{op} signifies the polarization resistance of the corrosion product on the surface of 316L or coating; R_{ct} represents the charge-transfer resistance of the double layer of 316L/solution or coating/solution; CPE_{dl} refers to the capacitance of the charge-transfer process. It can be seen that the R_{ct} of HEA coatings is

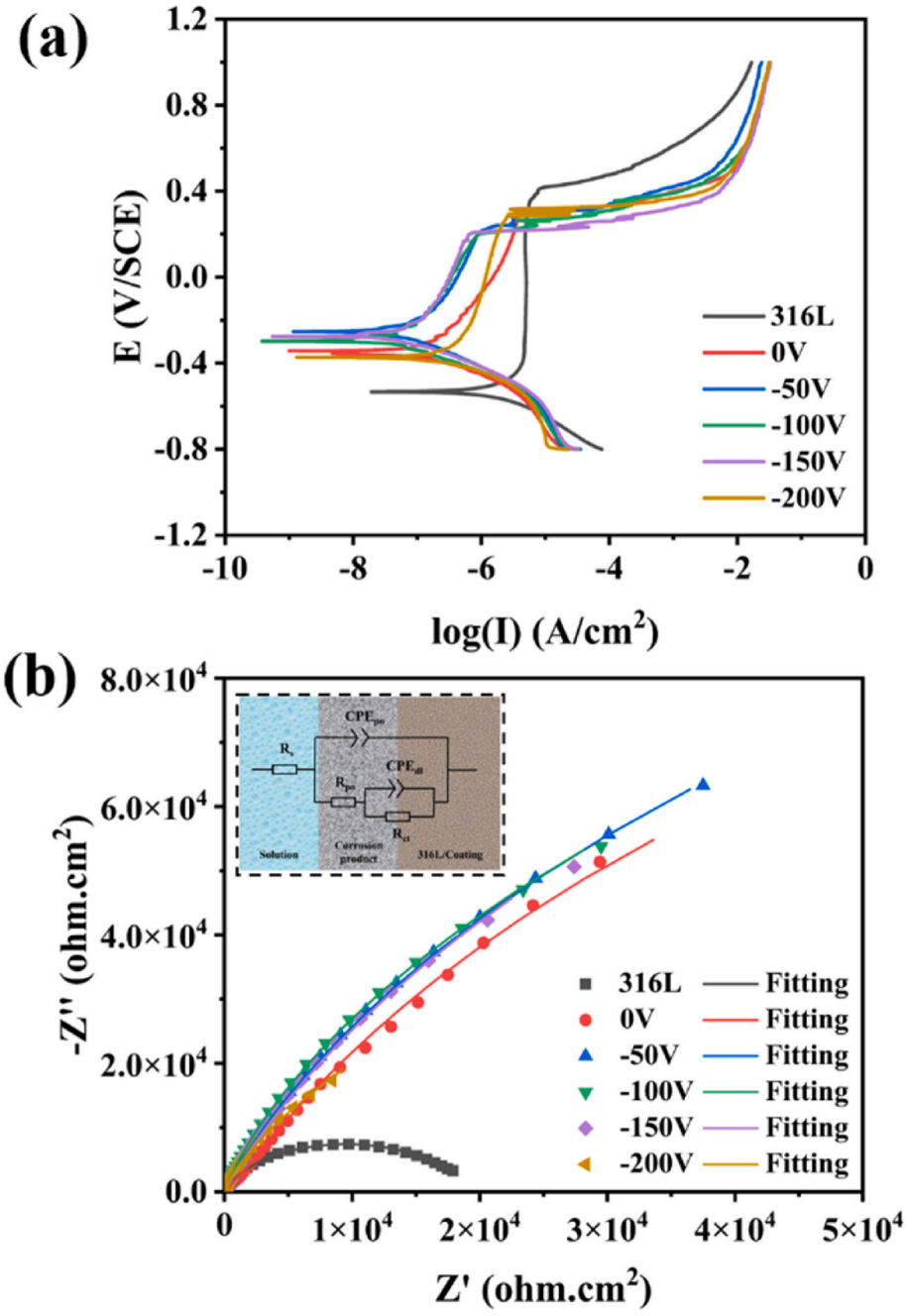


Fig. 11. (a) Potentiodynamic polarization curves and (b) Nyquist plot of FeCrAlTiSi coatings prepared under different biases and bare 316L substrate in a 3.5 wt% NaCl aqueous solution.

Table 3
Electrochemical corrosion parameter of FeCrAlTiSi coatings under different biases and bare 316L substrate from the polarization curves in Fig. 11(a) through Tafel extrapolation and Stern-Geary Equation.

Sample	E_{corr}(V)	I_{corr}(μA · cm^{-2})	β_a(mV•dec^{-1})	β_b(mV•dec^{-1})	R_p(kΩ)
316L	−0.534	1.409	227.78	127.11	25.14
0V	−0.345	0.122	254.98	119.49	289.65
−50V	−0.253	0.043	158.91	110.78	661.06
−100V	−0.280	0.025	115.14	97.43	909.29
−150V	−0.265	0.037	147.06	102.07	701.46
−200V	−0.370	0.225	266.45	109.45	150.01

between 269.75 and 309.67 kΩ · cm^2, about one order of magnitude larger than that of 17.254 kΩ · cm^2 for 316L. Specifically, the FeCrAlTiSi HEA coating at −100V has the largest R_{ct} indicating exceptional corrosion resistance, in good agreement with the potentiodynamic polarization analysis.

The superior corrosion resistance of the FeCrAlTiSi amorphous coatings to 316L could be attributed to the combined beneficial effect of chemical composition, the structural and compositional homogeneity, and the intrinsic amorphous structure. In terms of chemical composition, in comparison with 316L (with major constituents of Fe, Cr, Ni, and Mo), the Al, Ti, and Si elements in our coatings possess much higher oxygen affinities as well as high thermodynamic stabilities of their respective oxides in aqueous solution. The preferential oxidation of Al, Ti, and Si to form Al_2O_3, TiO_2, and SiO_2 has been verified by our XPS

Table 4
EIS equivalent circuit parameters for FeCrAlTiSi coatings prepared under different biases and bare 316L substrate in a 3.5 wt% NaCl aqueous solution.

Sample	R_s ($\Omega \cdot cm^2$)	R_{Po} ($k\Omega \cdot cm^2$)	CPE_{po} ($10^{-5}F \cdot cm^{-2}$)	n	R_{ct} ($k\Omega \cdot cm^2$)	CPE_{dl} ($10^{-5}F \cdot cm^{-2}$)	n
316L	23.15	1.932	2.26	0.878	17.254	1.02	0.729
0V	23.72	18.667	10.20	0.784	269.750	2.62	0.818
−50V	25.48	24.376	11.00	0.884	298.950	2.24	0.609
−100V	26.88	25.302	15.00	0.902	309.670	1.94	0.501
−150V	24.57	23.693	16.40	0.869	295.010	2.85	0.627
−200V	23.61	18.537	49.30	0.899	267.590	8.84	0.510

results (Fig. 7). The role of the structural and compositional homogeneity is obvious upon considering the polycrystal structure of 316L and the amorphous structure of FeCrAlTiSi coatings, which may be further verified by the very low I_{corr} registered in the active zone before passivation in potentiodynamic polarization curve (Fig. 11a). Besides the composition and homogeneity perspective, the intrinsic amorphous structure is beneficial for the rapid formation of electrochemically stable and compact passive film on the surface of FeCrAlTiSi coating, as reported that the growth mechanism of passive film is sensitive to the amorphous structure since amorphous metallic coatings, unlike crystalline materials, are thermodynamically metastable with higher energy, therefore, the atoms in metallic glasses are electrochemically more active leading to faster passivation [43,61,62].

4. Conclusions

FeCrAlTiSi high-entropy alloy coatings had been fabricated through the combined technique of magnetron sputtering and multiarc ion plating and investigated through various techniques in terms of microstructure, mechanical properties, and electrochemical corrosion behavior. It was found that FeCrAlTiSi HEA coatings were in an amorphous state with a columnar structure. The substrate bias had a significant influence on the coatings' microstructure and comprehensive properties. As the substrate bias increased from 0 to −200 V, the coating thickness increased at first and then decreased while the average particle size followed the opposite trends, reaching the maximum and minimum values of 4.92 μm and 168.9 nm at −100 V, respectively. Correspondingly, the nanohardness and Young's modulus reached the maximum value of 16.04 GPa and 241.3 GPa, respectively, at −100 V as a result of the densest microstructure, high entropy effect, and the enhanced ion bombardment on the coating. Moreover, the FeCrAlTiSi HEA coatings exhibited superior electrochemical corrosion resistance in 3.5 wt% NaCl aqueous solution to that of 316 L stainless steel, as high as 60 times improvement at −100 V in terms of corrosion current density (I_{corr}) and charge transfer resistance (R_{ct}), owing to the compositional and structural homogeneity, the intrinsic amorphous structure, and the preferential formed and chemically stable oxide film predominantly composed of Al_2O_3, TiO_2, and SiO_2.

CRediT authorship contribution statement

Li Zuo: Writing – original draft. **Qi Gao:** Data curation. **Yi-Cun Xu:** Conceptualization. **Zuo-Ming Xie:** Investigation. **Lin-Chao Zhang:** Data curation. **Rui Liu:** Conceptualization. **Li-Long Pang:** Writing – review & editing. **Zhi-Guang Wang:** Methodology, Conceptualization. **Gai-Ling Lu:** Formal analysis, Data curation. **Jun-Feng Yang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Nos. U22B2066,12064044,52273320), National Key Research and Development Program of China (Grant Nos. 2022YFE03140002, 2023YFA1609000), Natural Science Foundation of Inner Mongolia (2023LHMS01009), the Major Science and Technology Projects of Anhui Province (No.202103a05020016), Natural Science Foundation and Innovation R&D Program of Anhui Province (No. 2308085J07, 202423i08050026), the open competition project to select the best candidates to undertake major science and key research projects of Tongling City, Anhui Province (No.202101JB002). A proportion of this work was supported by the Academician workstation of Hangzhou Xingyu Carbon Environmental Tech Co., Ltd, and the Hefei Institutes of Physical Science Director's Fund Director's Fund (No.YZJJ-GGZX-2022-01,YZJJQY202306).

Data availability

Data will be made available on request.

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